

(2)

19 A

$$\tau = qE \times d = (2 \times 1.6 \times 10^{-19}) \left(\frac{2.0 \times 10^2}{0.001} \right) (4.0 \times 10^{-12} \sin 30^\circ)$$
$$= 1.3 \times 10^{-25} \text{ Nm}$$

20 D An α -particle has mass $4u$ and carries charge $+2e$.